Exploring DoRA: Weight-Decomposed Low-Rank Adaptation Across Modalities

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Introduction

Fine-tuning large pre-trained models is prohibitively expensive when all parameters must be updated. **LoRA** [2] addresses this by freezing model weights W_0 and injecting trainable low-rank matrices $\Delta W = BA$, reducing trainable parameters to roughly 1% of total. However, LoRA couples magnitude and directional updates within the same low-rank perturbation, which can introduce gradient interference especially on small datasets.

DoRA [1] decomposes each weight matrix into a learnable magnitude scalar m and a unit-norm direction updated via LoRA: $W' = m \cdot (W_0{+}BA)/\|W_0{+}BA\|_c$. The overhead is $+d$ scalars per layer ($< 0.1\%$ of parameters). The paper reports consistent gains over LoRA on NLP benchmarks at equivalent parameter budgets. Our project re-implements DoRA from scratch in PyTorch and evaluates it across four modalities: NLP (RoBERTa/GLUE), audio (Wav2Vec2/Speech Commands), vision (ViT and SigLIP/Cornell Grasp), and robotics (SmolVLM/Push-T). The paper we reproduce is *DoRA: Weight-Decomposed Low-Rank Adaptation* by Liu et al. (2024).

Chosen Result

We target DoRA's primary claim from Table 1: at comparable rank and parameter budget, DoRA can outperform LoRA on LLaMA-family commonsense reasoning benchmarks. We evaluate the same LoRA-vs.-DoRA claim on smaller GLUE classification tasks (SST-2, MRPC, RTE), then extend it to audio classification, visual grasp pose regression, and VLA action prediction.

Methodology

Custom DoRA implementation. We implement `DoRALinear`, a drop-in replacement for `nn.Linear` storing frozen W_0, low-rank A, B (init so $BA{=}0$), and magnitude m initialized to column norms of W_0. A parallel `LoRALinear` baseline (identical minus magnitude) enables fair ablation. Adapter state is saved separately from frozen weights.

NLP — GLUE (RoBERTa-base, 125M). Target modules: `query, key, value` projections. Rank $r = 8$, $\alpha = 16$, AdamW, cosine LR ($2{\times}10^{-4}$ for adapters, $2{\times}10^{-5}$ for head), 5 epochs, bfloat16. Tasks: SST-2 (67k, sentiment), MRPC (3.7k, paraphrase), RTE (2.5k, entailment). We also run a rank sweep ($r \in \{2, 4, 8, 16, 32\}$) on MRPC and a scale study with TinyLlama-1.1B and OpenLLaMA-3B.

Audio — Speech Commands (Wav2Vec2-base, 67M). 12-class keyword spotting (KWS-12). Same adapter config; feature encoder frozen; sequence classification head trained. 5 epochs, batch size 8, lr $2{\times}10^{-4}$; 84.8k train / 10k val / 4.9k test utterances.

Vision — Cornell Grasp (ViT-B/16 and SigLIP-B/16, ∼87–93M). Task: predict 6D grasp pose $(x, y, \sin 2\theta, \cos 2\theta, w, h)$ from a 640×480 RGB image. Metric: Cornell success rate — IoU ≥ 0.25 *and* $|\Delta\text{angle}| \leq 30$. 885 images (708 train / 177 val), image-level split. Trained for 30 epochs. Comparison: DoRA vs. LoRA vs. full fine-tuning.

VLA — Push-T (SmolVLM-256M). Task: predict 2D end-effector action (x, y) from RGB frame + text instruction. DoRA and LoRA applied to visual attention projections; trained for 3 epochs. Metric: action MSE on held-out rollouts.

Results & Analysis

NLP — GLUE

DoRA supports the paper's broader LoRA-vs.-DoRA claim on our low-data GLUE tasks: +0.3 pp on RTE, +0.6 pp F1 on MRPC vs. LoRA. On SST-2 (large data), LoRA slightly edges DoRA, consistent with the paper's finding that magnitude decoupling provides no benefit when gradients are already well-conditioned by data abundance. **The most striking result is full fine-tuning's collapse**:

Task (train size)	Metric	DoRA	LoRA	Full FT	Trainable
SST-2 (67k)	Accuracy	93.1%	**93.3%**	93.2%	~1%
RTE (2.5k)	Accuracy	**71.1%**	70.8%	54.9%	~1%
MRPC (3.7k)	F1	**90.7%**	90.1%	81.2%	~1%
MRPC (3.7k)	Accuracy	**87.0%**	85.8%	68.4%	~1%

Table 1: GLUE results. RoBERTa-base, rank=8. DoRA/LoRA outperform full FT on small tasks.

54.9% on RTE (near-chance) and 68.4% accuracy on MRPC, confirming that updating all 125M parameters on 2.5k examples causes catastrophic overfitting. The rank sweep on MRPC shows DoRA at r=16 (90.3% combined) matching LoRA at r=32 (90.1%), demonstrating parameter efficiency.

Audio — Speech Commands

DoRA achieves **98.6% validation / 89.7% test accuracy** vs. LoRA's 98.5% / 89.0%, and trains in **183.6 min** vs. 190.8 min (10% speedup) despite the magnitude scalar overhead. Most notably, DoRA's average train loss is **44.6%** less and convergence is faster.

Vision — Cornell Grasp

Backbone	Method	Success Rate	Trainable
ViT-B/16	DoRA	7.9%	~1%
ViT-B/16	LoRA	6.2%	~1%
ViT-B/16	Full FT	0.6%	100%
SigLIP-B/16	DoRA	19.8%	~1%
SigLIP-B/16	LoRA	**20.3%**	~1%
SigLIP-B/16	Full FT	15.8%	100%

Table 2: Cornell Grasp success rate (IoU $\geq$0.25 and $|\Delta\theta| \leq 30$).

Results diverge from the NLP trend: **DoRA does not consistently outperform LoRA**. On ViT-B/16, DoRA wins (+1.7 pp); on SigLIP-B/16, LoRA edges DoRA (20.3% vs. 19.8%). Full fine-tuning collapses on both backbones (Cornell: 885 images). The dominant factor is *backbone quality*: SigLIP achieves 3× better success than ViT, a gap driven by pretraining rather than adapter method.

VLA — Push-T

Both methods show rapid MSE reduction (DoRA: 63.6 → 27.9; LoRA: 62.1 → 24.8). **LoRA outperforms DoRA** here — with only 3 epochs and 256M parameters, magnitude decoupling has too few updates to provide directional benefit.

Reflections

Key takeaways. DoRA's gains are *task- and modality-dependent*: largest on low-data NLP, minimal or absent on vision and VLA. Across all modalities, adapters (DoRA and LoRA) vastly outperform full fine-tuning on small datasets — a finding more robust than DoRA's narrow edge over LoRA.

Unexpected findings. Per-epoch tracking reveals **magnitude drift is near-zero** (relative drift $<$ 0.7% across all runs). We hypothesize bfloat16 precision causes magnitude gradients to fall below the quantization step, freezing m at its pretrained value. DoRA's advantage likely comes from the unit-norm directional constraint, not magnitude learning — suggesting DoRA and LoRA are closer to equivalent than the paper (float32) reports.

Challenges. (1) Cornell Grasp (885 images) had high run variance. (2) VLA memory constraints limited us to 3 epochs with SmolVLM-256M vs. 7B OpenVLA. (3) GLUE reproduction required careful LR schedule, per-component weight decay, and BF16 matching.

Future work. Full VLA training with 4-bit quantization; rank-adaptive DoRA (scale r by magnitude drift per layer); float32 re-run to quantify bfloat16's masking effect.

References

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